

Course Objectives:

1. To provide a comprehensive understanding of the subject matter.
2. To develop critical thinking and analytical skills.
3. To enhance the ability to apply theoretical concepts to real-world scenarios.
4. To encourage independent learning and self-motivation.
5. To foster effective communication and collaboration skills.
6. To prepare students for further academic or professional pursuits in the field.
7. To promote ethical and responsible behavior in the discipline.

1. Chemistry of Atomic and Molecular Structure

- Atomic structure refers to the fundamental makeup of an atom, including its nucleus and electrons.
- The nucleus of an atom is composed of protons and neutrons, which are collectively referred to as nucleons.
- The number of protons in the nucleus of an atom is known as its atomic number, and it determines the element to which the atom belongs.
- Electrons are negatively charged particles that orbit the nucleus of an atom in shells or energy levels.
- The arrangement of electrons in an atom's outermost shell, known as its valence shell, determines its chemical properties and reactivity.
- A molecule is a group of two or more atoms held together by chemical bonds.
- The type and strength of the chemical bonds between atoms in a molecule determine its shape, properties, and reactivity.
- Understanding the chemistry of atomic and molecular structure is essential for predicting and explaining chemical reactions and the properties of substances.

Advanced Materials

Advanced materials refer to materials that have been developed through the incorporation of new or improved properties, making them superior to traditional materials in terms of performance or sustainability. Some examples of advanced materials include liquid crystals, nanomaterials, graphite and fullerenes, and green chemistry.

Liquid Crystals

- Liquid crystals are a state of matter that has properties between those of conventional liquids and those of solid crystals.
- They can flow like a liquid, but have the long-range order of a solid crystal.

- Liquid crystals are used in a variety of applications, including displays, sensors, and thermometers.

Nanomaterials

- Nanomaterials are materials with at least one dimension in the nanometer scale (1-100 nm).
- They exhibit unique properties due to their small size and high surface area to volume ratio.
- Nanomaterials are used in a variety of applications, including electronics, medicine, and energy.

Graphite and Fullerenes

- Graphite is a form of carbon that consists of layers of carbon atoms arranged in a hexagonal lattice.
- Fullerenes are a family of carbon allotropes that are composed entirely of carbon, in the form of a hollow sphere, ellipsoid, or tube.
- Graphite and fullerenes have unique properties, including high electrical conductivity and strength, and are used in a variety of applications, including batteries and lubricants.

Green Chemistry

- Green chemistry is the design of chemical products and processes that reduce or eliminate the use or generation of hazardous substances.
- It aims to minimize the environmental impact of chemical processes and products, while maintaining their efficiency and effectiveness.
- Green chemistry is used in a variety of applications, including the development of safer and more sustainable products and processes.

2. To enable the students to understand and apply the detailed concepts of spectroscopic techniques:

- Spectroscopy is the study of the interaction between matter and electromagnetic radiation.
- Spectroscopic techniques are used to analyze the chemical composition and structure of materials.
- Some common spectroscopic techniques include infrared spectroscopy, ultraviolet-visible spectroscopy, and nuclear magnetic resonance spectroscopy.
- Infrared spectroscopy measures the absorption of infrared radiation by molecules, providing information about their chemical bonds and functional groups.
- Ultraviolet-visible spectroscopy measures the absorption of ultraviolet and visible light by molecules, providing information about their electronic

structure.

- Nuclear magnetic resonance spectroscopy measures the interaction of nuclear spins with an applied magnetic field, providing information about the chemical environment of the nuclei.
- These techniques can be used to identify unknown compounds, determine the purity of a sample, and study the reaction mechanisms of chemical reactions.
- To apply these techniques, students must have a strong understanding of the underlying principles and be able to interpret the resulting spectra.

Stereochemistry

Stereochemistry is the branch of chemistry that deals with the spatial arrangement of atoms and molecules and their effect on the physical and chemical properties of substances.

- Stereoisomers are isomers that have the same molecular formula and sequence of bonded atoms, but differ in the three-dimensional orientation of their atoms in space.
- There are two main types of stereoisomers: enantiomers and diastereomers.
- Enantiomers are mirror images of each other and are non-superimposable. They have the same physical and chemical properties, except for their ability to rotate plane-polarized light in opposite directions.
- Diastereomers are stereoisomers that are not mirror images of each other. They have different physical and chemical properties.
- The study of stereochemistry is important in the synthesis and identification of compounds, as well as in understanding their biological activity.

3. To enable the students to understand and apply the concepts related to Electrochemistry, Batteries, Corrosion

Electrochemistry is the branch of chemistry that deals with the relationship between electricity and chemical reactions. It involves the study of the movement of electrons in chemical reactions and the resulting changes in matter.

Some key concepts in electrochemistry include:

- **Redox reactions:** These are reactions in which electrons are transferred from one species to another. The species that loses electrons is oxidized, while the species that gains electrons is reduced.
- **Electrochemical cells:** These are devices that convert chemical energy into electrical energy or vice versa. There are two types of electrochemical cells: galvanic cells and electrolytic cells.
- **Galvanic cells:** These are electrochemical cells in which a spontaneous redox reaction generates an electric current. A common example is a

battery.

- **Electrolytic cells:** These are electrochemical cells in which an electric current drives a non-spontaneous redox reaction. A common example is the electrolysis of water to produce hydrogen and oxygen gas.

Batteries are devices that store chemical energy and convert it into electrical energy. There are many different types of batteries, but they all consist of one or more electrochemical cells. Some common types of batteries include:

- **Primary batteries:** These are batteries that cannot be recharged. Once the chemical reactions inside the battery are complete, the battery is dead and must be disposed of.
- **Secondary batteries:** These are batteries that can be recharged. The chemical reactions inside the battery can be reversed by applying an external electric current.

Corrosion is the degradation of a material due to chemical reactions with its environment. It is a common problem in many industries, as it can cause damage to infrastructure and equipment. Some common causes of corrosion include:

- **Oxidation:** This is the most common form of corrosion. It occurs when a metal reacts with oxygen to form an oxide.
- **Electrochemical corrosion:** This occurs when two different metals are in contact with each other in the presence of an electrolyte. The more reactive metal will corrode, while the less reactive metal will be protected.
- **Galvanic corrosion:** This is a type of electrochemical corrosion that occurs when two different metals are in contact with each other in the presence of an electrolyte. The more reactive metal will corrode, while the less reactive metal will be protected.

In summary, electrochemistry, batteries, and corrosion are important concepts in chemistry that have many practical applications. Understanding these concepts can help students to better understand the world around them and to make informed decisions about the use of technology.

Chemistry of Engineering Materials: Cement

Cement is a binding material that is used in construction to set, harden, and adhere to other materials. The chemical composition of cement depends on the raw materials used in its manufacturing. These raw materials are lime, silica, alumina, and iron oxide.

Portland cement, the most common type of cement, is made up of four main compounds: tricalcium silicate ($3\text{CaO} \cdot \text{SiO}_2$), dicalcium silicate ($2\text{CaO} \cdot \text{SiO}_2$), tricalcium aluminate ($3\text{CaO} \cdot \text{Al}_2\text{O}_3$), and a tetra-calcium aluminoferrite ($4\text{CaO} \cdot \text{Al}_2\text{O}_3\text{Fe}_2\text{O}_3$).

Cement can be used alone, but its normal use is in mortar and concrete, where it is mixed with inert material known as aggregate. Mortar is cement mixed with sand or crushed stone that must be less than approximately 5 mm in size.

With a better understanding of cement chemistry, scientists could potentially alter production or change ingredients so that concrete has less of an impact on emissions, or add ingredients that are capable of actively absorbing carbon dioxide.

4. To enable the students to understand and apply detailed concepts of water source, water impurities, hardness

Water is an essential resource for life and is used for various purposes such as drinking, agriculture, and industrial processes. It is important to understand the sources of water, the impurities present in it, and its hardness to ensure its safe and efficient use.

1. **Water sources:** Water can be obtained from various sources such as surface water (rivers, lakes, and reservoirs), groundwater (wells and springs), and rainwater. The choice of water source depends on factors such as availability, quality, and cost.
2. **Water impurities:** Water from natural sources is not pure and contains various impurities such as dissolved minerals, organic matter, and microorganisms. These impurities can affect the taste, odor, and color of water and can also have health implications.
3. **Hardness:** Hardness is a measure of the concentration of calcium and magnesium ions in water. Hard water can cause scaling in pipes and appliances and can also interfere with the action of soap and detergents. Water hardness can be reduced by methods such as ion exchange and reverse osmosis.

In conclusion, understanding the concepts of water source, water impurities, and hardness is important for the safe and efficient use of water. Students should be able to apply these concepts to make informed decisions about water use and treatment.

Water and Boiler Troubles Used in Industry as well as Analysis of Coal & Determination of Calorific Values

1. Coal boilers use small amounts of water for boiler blowdown. In this process, water is bled from the boiler to get rid of impurities that accumulate and form sludge that can impair a plant's performance.
2. The pH value of the boiler water is a number between zero and fourteen. Values below seven are acidic, seven is neutral, and values above seven

are alkaline. The pH factor influences scale formation and the corrosive tendencies of boiler water.

3. Most coal-fired power station boilers use pulverized coal, and many of the larger industrial water-tube boilers also use this pulverized fuel. This technology is well developed, and there are thousands of units around the world, accounting for well over 90% of coal-fired capacity.
4. The coal is ground (pulverized) to a fine powder, so that less than 2% is +300 micro meter (μm) and 70-75% is below 75 microns, for a bituminous coal. It should be noted that too fine a powder is wasteful of grinding mill power. On the other hand, too coarse a powder does not burn completely in the combustion chamber and results in higher unburnt losses.
5. The main heat losses of the boiler are as follows: incomplete combustion, excess air, flue gas heat loss due to water vapor in flue gas temperature, fuel type, boiler load, the pollution of surface heat losses from the boiler heating surface.
6. Moisture in coal is needed for determining the calorific (heating) value and handling properties of a coal. The moisture value given for a proximate analysis is the moisture measured as mass lost from a sample under specified conditions after heating in a moisture oven to 104 to 110°C.
7. Industrial boilers are used to produce steam or heat water for space and process heating and for the generation of mechanical power and electricity.

5. To enable the students to understand detailed concepts related to polymers, Polymerization, Polymer Blends

Polymers are large molecules composed of repeating structural units, typically connected by covalent chemical bonds. The process of converting monomers into polymers is called polymerization.

There are two main types of polymerization: addition polymerization and condensation polymerization. In addition polymerization, monomers react to form a polymer without the loss of any byproducts. In condensation polymerization, monomers react with the loss of a small molecule, such as water, to form a polymer.

Polymer blends are mixtures of two or more polymers. These blends can be used to create materials with unique properties that cannot be achieved by using a single polymer. Polymer blends can be classified into two types: miscible and immiscible. Miscible blends are homogeneous mixtures, while immiscible blends are heterogeneous mixtures.

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Polymer Composites

Polymer composites are materials made by combining a polymer matrix with a reinforcing material. The reinforcing material can be in the form of fibers, particles, or flakes, and is typically made of glass, carbon, or other materials. The polymer matrix can be a thermoplastic or thermosetting polymer.

Some key points to consider when studying polymer composites include:

1. The properties of the composite material depend on the properties of the individual components, as well as the way they are combined.
2. The reinforcing material provides strength and stiffness to the composite, while the polymer matrix provides toughness and ductility.
3. The interface between the reinforcing material and the polymer matrix is critical to the performance of the composite.
4. Polymer composites can be tailored to have specific properties by varying the type, amount, and orientation of the reinforcing material.
5. Polymer composites are used in a wide range of applications, including aerospace, automotive, construction, and sports equipment.

Content Contact

Content contact refers to the interaction between the content of a message and its recipient. It is an important aspect of communication as it determines how the message is received and interpreted by the recipient. Here are some key points to consider when it comes to content contact:

1. Clarity: The content of the message should be clear and easy to understand. This can be achieved by using simple language, avoiding jargon, and organizing the content in a logical manner.
2. Relevance: The content should be relevant to the recipient. This means that it should address their needs, interests, or concerns.
3. Accuracy: The content should be accurate and based on facts. This helps to establish credibility and trust with the recipient.
4. Tone: The tone of the content should be appropriate for the intended audience. For example, a formal tone may be appropriate for a business communication, while a more casual tone may be appropriate for a personal communication.
5. Length: The length of the content should be appropriate for the intended audience and the medium of communication. For example, a long-form article may be appropriate for a website, while a shorter message may be more appropriate for a text message or social media post.
6. Visuals: The use of visuals, such as images or videos, can help to enhance the content and make it more engaging for the recipient.

In summary, content contact is an important aspect of communication that should be carefully considered when crafting a message. By paying attention to factors such as clarity, relevance, accuracy, tone, length, and visuals, you can help to ensure that your message is well-received and effectively communicates your intended message.

Hours

- An hour is a unit of time equal to 60 minutes or 3,600 seconds.
- It is one of the seven base units in the International System of Units (SI).
- The symbol for hour is “h”.
- The hour is commonly used to measure the duration of events, to express the time of day, and to measure the amount of time spent on a task.
- The concept of the hour has been used since ancient times, with the earliest known reference to the hour being in the writings of the ancient Egyptians.
- The hour is divided into 60 minutes, and each minute is divided into 60 seconds.
- The hour is also used to measure angles, with one hour being equal to 15 degrees.
- The hour is used in both the 12-hour and 24-hour clock systems.
- In the 12-hour clock system, the day is divided into two 12-hour periods, with the first period running from midnight to noon and the second period running from noon to midnight.
- In the 24-hour clock system, the day is divided into 24 hours, with each hour being numbered from 0 to 23.
- The 24-hour clock system is commonly used in military, aviation, and other fields where precise timekeeping is essential.

Unit-1: 8

1. The number 8 is an even integer.
2. It is the cube of 2.
3. It is a composite number, with its divisors being 1, 2, 4, and 8.
4. In the decimal number system, 8 is the highest digit.
5. The number 8 is considered lucky in some cultures, such as Chinese culture.
6. In music, an octave is the interval between one musical pitch and another with double its frequency.
7. In computer science, a byte is typically composed of 8 bits.
8. In sports, the number 8 is often associated with the center position in various team sports.

Atomic and Molecular Structure: Molecular orbitals of diatomic molecules, Bond Order

Molecular orbitals are formed by the combination of atomic orbitals of two or more atoms. In diatomic molecules, the atomic orbitals of the two atoms combine to form molecular orbitals. These molecular orbitals can be bonding, anti-bonding, or non-bonding.

Bond order is a measure of the strength of a chemical bond. It is defined as the difference between the number of bonding electrons and the number of anti-bonding electrons, divided by two. A higher bond order indicates a stronger chemical bond.

In diatomic molecules, the bond order can be calculated using the molecular orbital diagram. The molecular orbitals are arranged in order of increasing energy, with the bonding orbitals lower in energy than the anti-bonding orbitals. The electrons are then placed in the molecular orbitals following the Aufbau principle, Hund's rule, and the Pauli exclusion principle.

For example, in the case of the oxygen molecule (O_2), the bond order is 2. This is because there are 8 valence electrons in the oxygen molecule, which are distributed among the molecular orbitals as follows: 2 electrons in the sigma 2s bonding orbital, 2 electrons in the sigma 2s* anti-bonding orbital, 4 electrons in the two pi 2p bonding orbitals, and 0 electrons in the sigma 2p bonding orbital. The bond order is then calculated as $(4-2)/2 = 2$.

In summary, the molecular orbitals of diatomic molecules are formed by the combination of atomic orbitals, and the bond order is a measure of the strength of the chemical bond. The bond order can be calculated using the molecular orbital diagram and the distribution of electrons among the molecular orbitals. A higher bond order indicates a stronger chemical bond.

Magnetic Characteristics and Numerical Problems

Magnetic characteristics refer to the properties of a material that determine its behavior in the presence of a magnetic field. Some of the important magnetic characteristics of materials are:

1. **Permeability:** It is the measure of the ability of a material to support the formation of a magnetic field within itself. It is represented by the symbol μ .
2. **Susceptibility:** It is the measure of how easily a material can be magnetized. It is represented by the symbol χ .
3. **Magnetic Moment:** It is a vector quantity that represents the magnetic strength and orientation of a magnet or other object that produces a

magnetic field.

4. **Magnetic Field Strength:** It is the measure of the magnetic field produced by a magnet or a current-carrying conductor. It is represented by the symbol H .
5. **Magnetic Flux Density:** It is the measure of the amount of magnetic flux passing through a unit area. It is represented by the symbol B .

Numerical problems involving magnetic characteristics typically involve the calculation of one or more of these quantities, given certain information about a magnetic system. For example, one might be asked to calculate the magnetic field strength at a certain point, given the current flowing through a conductor and its distance from the point in question. Another common type of problem involves calculating the magnetic force between two magnets, given their magnetic moments and the distance between them.

It is important to have a good understanding of the basic concepts and relationships between these magnetic characteristics in order to solve numerical problems effectively. Practice and familiarity with common problem-solving techniques can also be helpful in this regard.

Chemistry of Advanced Materials:

1. Advanced materials are materials that exhibit superior properties compared to traditional materials.
2. These properties can include strength, durability, conductivity, and reactivity.
3. The chemistry of advanced materials involves the study of the composition, structure, and properties of these materials.
4. This field of study also includes the development of new materials and the improvement of existing materials.
5. Some examples of advanced materials include nanomaterials, biomaterials, and smart materials.
6. Nanomaterials are materials that have at least one dimension in the nanometer range.
7. Biomaterials are materials that are used in medical applications, such as implants and prosthetics.
8. Smart materials are materials that can change their properties in response to external stimuli, such as temperature or pressure.
9. The development of advanced materials requires a deep understanding of the underlying chemistry and the ability to manipulate the composition and structure of materials at the molecular level.
10. The field of advanced materials chemistry is interdisciplinary, involving collaboration between chemists, physicists, materials scientists, and engineers.

Liquid Crystals

Introduction

Liquid crystals (LC) are liquids featuring a certain level of orientational order. Specifically, molecules in LCs tend to point to a certain direction, while they still have translational (positional) freedom . Liquid crystal phases represent a unique state of matter characterized by both mobility and order on a molecular and at the supramolecular levels .

Types

Liquid crystals can be classified into two main categories: thermotropic liquid crystals, and lyotropic liquid crystals. These two types of liquid crystals are distinguished by the mechanisms that drive their self-organization, but they are also similar in many ways . Thermotropic liquid crystals are obtained by melting a crystalline solid . Lyotropic liquid crystals are found in certain colloidal solutions, such as aqueous solutions of tobacco mosaic virus and certain polymers .

Applications

Liquid crystals have found a wide range of commercial applications. They are used, for example, in the liquid crystal displays (LCDs) in digital watches, calculators, and computer and video displays . Two groups of potential applications are the use of liquid crystals as anisotropic solvent for the determination of molecular properties by means of spectroscopy, and their use in analytical chemistry, particularly in gas chromatography . Liquid crystals are also an essential part of all life forms. Lyotropic liquid crystals are essential organic substances, DNA, lipids of cellular membranes and proteins are some examples of well known liquid crystals .

Industrial Applications

Liquid crystals have found a wide range of industrial applications, particularly in the field of display technology. Liquid crystal displays (LCDs) are widely used in various devices such as digital watches, calculators, and computer and video displays .

Important materials used as liquid crystals

Liquid crystals are substances that blend the structures and properties of the normally disparate liquid and crystalline solid states. They can be divided into 3 main types: thermotropic, lyotropic, and metallotropic. Thermotropic and lyotropic liquid crystals consist mostly of organic molecules, although a few minerals are also known.

The amphiphilic molecules generally used in the formation of lyotropic liquid crystals can be long-chain alcohols, deoxyribonucleic acid (DNA), gelatin, surfactants, and membrane phospholipids, whereas, the hydrophilic solvent is usually water .

Many biological materials form liquid crystals. Myelin, a fatty material extracted from nerve cells, was the first intensively studied liquid crystal. The tobacco mosaic virus, with its rodlike shape, forms a nematic phase.

Liquid crystals can be first organized into thermotropic (temperature dependent) and lyotropic (concentration dependent). Within thermotropic liquid crystals there are high molar mass materials (suitable for polymer) and low molar mass materials.

Graphite and Fullerene; Introduction, Structure and applications

Graphite and fullerene are two allotropes of carbon. Graphite has a layered structure in which hexagonal rings are connected to each other. The fullerene structure is similar to graphite, but a pentagonal ring may also exist in the fullerene structure. The most abundant form of fullerene is Buckminster fullerene (C60), which has 60 carbon atoms arranged in a spherical structure.

Graphene is a single layer of graphite and has properties that make it useful in electronics and composites. Like graphite, graphene conducts electricity well because it has delocalised electrons that are free to move across its surface. Fullerenes have a similar structure to graphite, which is made up of a sheet of connected hexagonal rings, but they have pentagonal (or sometimes heptagonal) rings that prevent the sheet from being planar. Buckyballs and buckytubes are terms used to describe them depending on their shape. Cylindrical fullerenes are referred to as nanotubes.

The discovery of C60 has led to a paradigm shift in the understanding of graphite, in particular graphene sheets on a small scale. It is now known that the most stable form of a carbon aggregate, containing tens to several thousands of atoms, is a closed buckyball or nanotube.

Nanomaterials; Introduction, Preparation, characteristics of nanomaterials and applications

Introduction

Nanomaterials are materials that have at least one dimension in the nanoscale range, typically between 1 and 100 nanometers. They can be classified as nano-objects or nanostructured materials. Nano-objects are discrete pieces of material, while nanostructured materials have their internal or surface structure in the nanoscale dimension. Nanomaterials can be of natural existence, artificially manufactured or incidentally formed.

Preparation

Nanomaterials can be produced with outstanding magnetic, electrical, optical, mechanical, and catalytic properties that are substantially different from their bulk counterparts. The nanomaterial properties can be tuned as desired via precisely controlling the size, shape, synthesis conditions, and appropriate functionalization.

Characteristics of nanomaterials

Nanomaterials have unique properties that are different from their bulk counterparts. These properties can be tuned by controlling the size, shape, synthesis conditions, and appropriate functionalization. Some of the properties that can be achieved with nanomaterials include outstanding magnetic, electrical, optical, mechanical, and catalytic properties.

Applications

Nanomaterials have become one of the most promising topics in different fields, such as physics, chemistry, engineering, and biology, with many breakthroughs that will change the direction of nanotechnology advances in a large range of applications.

Nanomaterials: Carbon Nano Tubes (CNT)

- Carbon nanotubes (CNTs) are cylinder-shaped allotropic forms of carbon, most widely produced under chemical vapor deposition .
- They possess astounding chemical, electronic, mechanical, and optical properties .
- Being among the most promising materials in nanotechnology, they are also likely to revolutionize medicine .
- Carbon nanomaterials are an extensive family of carbon allotropes, consisting of 0-dimensional fullerenes and quantum dots, 1-dimensional carbon nanotubes (CNTs), 2-dimensional graphene, and 3-dimensional nanodiamonds and nanohorns .
- Carbon nanomaterials are used in a broad range of applications due to their unique physical and chemical properties .
- Single-walled carbon nanotubes (SWCNTs), which can be considered as seamless cylinders of graphene, have been at the forefront of nanotechnology research for the past two decades .
- They possess a range of exceptional properties including high strength (37 GPa), thermal conductivity (3500 W/m/K), and ballistic electronic transport .
- A carbon nanotube(CNT) is a tube made of carbon with diameters typically measured in nanometers .

- Single-wall carbon nanotubes(SWCNTs) are one of the allotropes of carbon, intermediate between fullerenecages and flat graphene, with diameters in the range of a nanometre .

Green Chemistry: Introduction, 12 principles and importance of green Synthesis, Green

Green chemistry, also known as sustainable chemistry, is the design of chemical products and processes that reduce or eliminate the use or generation of hazardous substances. Green chemistry applies across the life cycle of a chemical product, including its design, manufacture, use, and ultimate disposal.

The 12 principles of green chemistry are:

1. Prevention: It is better to prevent waste than to treat or clean up waste after it has been created.
2. Atom Economy: Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product.
3. Less Hazardous Chemical Syntheses: Wherever practicable, synthetic methods should be designed to use and generate substances that possess little or no toxicity to human health and the environment.
4. Designing Safer Chemicals: Chemical products should be designed to effect their desired function while minimizing their toxicity.
5. Safer Solvents and Auxiliaries: The use of auxiliary substances (e.g., solvents, separation agents, etc.) should be made unnecessary wherever possible and innocuous when used.
6. Design for Energy Efficiency: Energy requirements of chemical processes should be recognized for their environmental and economic impacts and should be minimized. If possible, synthetic methods should be conducted at ambient temperature and pressure.
7. Use of Renewable Feedstocks: A raw material or feedstock should be renewable rather than depleting whenever technically and economically practicable.
8. Reduce Derivatives: Unnecessary derivatization (use of blocking groups, protection/ deprotection, temporary modification of physical/chemical processes) should be minimized or avoided if possible, because such steps require additional reagents and can generate waste.
9. Catalysis: Catalytic reagents (as selective as possible) are superior to stoichiometric reagents.
10. Design for Degradation: Chemical products should be designed so that at the end of their function they break down into innocuous degradation products and do not persist in the environment.
11. Real-time analysis for Pollution Prevention: Analytical methodologies need to be further developed to allow for real-time, in-process monitoring and control prior to the formation of hazardous substances.
12. Inherently Safer Chemistry for Accident Prevention: Substances and the form of a substance used in a chemical process should be chosen to min-

imize the potential for chemical accidents, including releases, explosions, and fires.

Green synthesis is the process of designing chemical reactions and processes that reduce or eliminate the use and generation of hazardous substances. It is an important aspect of green chemistry as it helps to minimize the environmental impact of chemical processes and products.

Green chemistry is important because it promotes sustainability and reduces the negative impact of chemical processes and products on the environment and human health. By designing chemical products and processes that are safer and more environmentally friendly, green chemistry helps to protect the planet and its inhabitants. It also has the potential to reduce costs and increase efficiency in the chemical industry.

Chemicals, Synthesis of typical organic compounds by conventional and Green route (Adipic)

Adipic acid is an organic compound that is typically synthesized through conventional and green routes. The conventional route involves the use of benzene as a starting material, while the green route uses glucose as a starting material .

The conventional route requires high pressure and temperature, while the green route requires less temperature and pressure .

Green techniques have been applied to synthesize both well-known chemical compounds by more sustainable routes and completely new materials. A range of nanosized materials and composites can be produced by greener routes, including nanoparticles of metals, non-metals, their oxides and salts, aerogels or quantum dots .

In organic chemistry, green techniques include reactions of C–H bond activation, fluororous, solid-supported, bio- and asymmetric catalysis and synthesis, use of water and other green solvents (in particular, ionic liquids (ILs)) or without any solvent, microwave-, ultrasound- and ultraviolet (UV)-assisted reactions, and use of flow reactors .

The ideology of Green Chemistry calls for the development of new chemical reactivities and reaction conditions that can potentially provide benefits for chemical syntheses in terms of resource and energy efficiency, product selectivity, operational simplicity, and health and environmental safety .

Acid and Paracetamol

- Mefenamic Acid + Paracetamol is a combination of two medicines: Mefenamic Acid and Paracetamol. It works by blocking the release of certain

chemical messengers that cause fever, pain and inflammation (redness and swelling).

- Paracetamol may be made by acetylation of para-aminophenol (obtained by reduction of para-nitrophenol) with acetic acid or acetic anhydride.
- Paracetamol is used as an analgesic and antipyretic drug. It is the preferred alternative analgesic-antipyretic to aspirin (acetylsalicylic acid), particularly in patients with coagulation disorders, individuals with a history of peptic ulcer or who cannot tolerate aspirin, as well as in children.

Environmental impact of Green chemistry on society

- Green chemicals either degrade to innocuous products or are recovered for further use. Plants and animals suffer less harm from toxic chemicals in the environment. Lower potential for global warming, ozone depletion, and smog formation. Less chemical disruption of ecosystems.
- Scientists and chemists can significantly minimize the risk to environment and health of human by the help of all the valuable ideology of green chemistry. The three main developments in green chemistry include use of super critical carbon dioxide water as green solvent, aqueous hydrogen peroxide as an oxidizing agent and use of hydrogen in asymmetric synthesis.
- Green chemistry is the design of high-performing, cost-effective technology that is safer for the environment and human health.
- In the Global Chemicals Outlook II it was concluded that the global goal to minimize adverse impacts of chemicals and waste will not be achieved by 2020. More urgent and ambitious worldwide action by stakeholders is required.

Unit-2: 8

1. I'm sorry, but I need more information to provide a response. Can you please clarify what you mean by "Unit-2: 8"?
2. Is this a chapter or section in a specific book or course?
3. What is the subject or topic of this unit?
4. Providing more information will help me generate a more accurate and helpful response. Thank you.

Spectroscopic Techniques and Applications: Elementary idea and simple applications of

Spectroscopy is the study of the interaction between matter and electromagnetic radiation. It is a powerful tool for the analysis of the structure and composition of materials. Some of the elementary ideas and simple applications of spectroscopy are:

1. **Identification of elements and compounds:** Spectroscopy can be used to identify the presence of specific elements or compounds in a sample by analyzing the characteristic absorption or emission spectra.
2. **Quantitative analysis:** Spectroscopy can be used to determine the concentration of a particular substance in a sample by measuring the intensity of its characteristic spectral lines.
3. **Molecular structure determination:** Spectroscopy can provide information about the structure of molecules, including bond lengths, bond angles, and the presence of functional groups.
4. **Remote sensing:** Spectroscopy can be used to remotely sense the composition of materials, such as the atmosphere of a planet or the surface of a distant object.

These are just a few examples of the many applications of spectroscopy. Spectroscopy is a versatile and powerful tool that is widely used in many fields, including chemistry, physics, biology, and environmental science.

UV, IR and NMR Spectroscopy

UV, IR and NMR spectroscopy are analytical techniques used to determine the structure of molecules.

UV Spectroscopy

- UV spectroscopy uses ultraviolet light to excite electrons in a molecule to higher energy levels.
- The absorption of UV light by a molecule is measured and used to determine the structure of the molecule.
- UV spectroscopy is commonly used to identify functional groups in organic molecules.

IR Spectroscopy

- IR spectroscopy uses infrared light to excite vibrations in the bonds of a molecule.
- The absorption of IR light by a molecule is measured and used to determine the structure of the molecule.
- IR spectroscopy is commonly used to identify functional groups in organic molecules.

NMR Spectroscopy

- NMR spectroscopy uses radio waves to excite the nuclei of atoms in a molecule.

- The absorption of radio waves by a molecule is measured and used to determine the structure of the molecule.
- NMR spectroscopy is commonly used to determine the connectivity of atoms in a molecule.

Numerical Problems

- Numerical problems in UV, IR and NMR spectroscopy involve using the principles of these techniques to solve problems related to the structure of molecules.
- These problems may involve calculating the absorption of light by a molecule, predicting the spectrum of a molecule, or interpreting the spectrum of a molecule to determine its structure.

Stereochemistry: Optical isomerism in compounds without chiral carbon, Geometrical

- Optical isomerism occurs when a compound has similar bonds but a different spatial arrangement of atoms, resulting in non-superimposable mirror images, known as optical isomers or enantiomers.
- Enantiomers differ from one another in their optical activity.
- Geometrical stereoisomers are a common occurrence in heteroleptic compounds. This type of isomerism occurs due to the different geometric configurations of the molecules.
- Optical isomerism occurs in chiral molecules in solution, which appear to be mirror images of one another.
- Optical isomers are named so because of their effect on plane-polarized light.
- Simple substances that show optical isomerism exist as two isomers known as enantiomers.

Isomerism and Chiral Drugs

Isomerism is the phenomenon where two or more compounds have the same molecular formula but different arrangements of atoms in space. This can result in different physical and chemical properties of the compounds. Isomerism can occur as a drug degradation pathway, where a drug gets converted into its optical or geometric isomers, a process known as isomerization.

Stereoisomers are isomeric pairs with identical atomic and molecular formulas but different three-dimensional arrangements. Many drugs available on the market exhibit optical isomerism, where a molecule cannot be superimposed on its mirror image. Such a molecule is called chiral, and each of these images is called an enantiomer.

The stereoisomeric composition of a drug with a chiral center should be known and the quantitative isomeric composition of the material used in pharmacologic,

toxicologic, and clinical studies. The importance of chiral drugs in the drug development space cannot be understated. In pharmaceutical industries, 56% of the drugs currently in use are chiral molecules and 88% of the last ones are marketed as racemates (or racemic mixtures), consisting of an equimolar mixture of two enantiomers.

It is important to note that the word chiral describes the three-dimensional architecture of the molecule and does not reveal the stereochemical composition. Hence “chiral drug” does not say whether the drug is racemic (racemic drug), single enantiomer (chiral specific drug) or some other combination of stereoisomers.

Unit-3: 8

1. Point 1
2. Point 2
3. Point 3
4. Point 4
5. Point 5
6. Point 6
7. Point 7
8. Point 8

Electrochemistry and Batteries: Basic concepts of electrochemistry

Electrochemistry is the branch of chemistry that deals with the relationship between electrical energy and chemical reactions. It involves the study of the movement of electrons in chemical reactions, and how this movement can be used to generate electricity or drive chemical reactions.

Some basic concepts in electrochemistry include:

1. **Redox reactions:** These are reactions in which electrons are transferred from one molecule to another. The molecule that loses electrons is oxidized, while the molecule that gains electrons is reduced.
2. **Electrochemical cells:** These are devices that convert chemical energy into electrical energy, or vice versa. They consist of two electrodes (an anode and a cathode) and an electrolyte, which conducts ions between the electrodes.
3. **Galvanic cells:** These are a type of electrochemical cell in which a spontaneous redox reaction generates an electric current. The anode is the electrode where oxidation occurs, while the cathode is the electrode where reduction occurs.
4. **Electrolysis:** This is the process of using an electric current to drive a non-spontaneous chemical reaction. In electrolysis, the anode is the

electrode where the electric current enters the cell, while the cathode is the electrode where the electric current leaves the cell.

5. **Batteries:** These are devices that store electrical energy in the form of chemical energy. They consist of one or more electrochemical cells connected in series or parallel. When the battery is connected to an external circuit, a chemical reaction occurs that generates an electric current.

These are some of the basic concepts of electrochemistry. Understanding these concepts is essential for studying the applications of electrochemistry, such as in batteries, fuel cells, and corrosion prevention.

Batteries; Classification and applications of Primary Cells (Dry Cell) and Secondary Cells

A battery is a combination of many cells, which can be divided into two types: primary and secondary cells.

- **Primary Cells:** A primary cell is a battery that is designed to be used only one time. It creates an electric current through an irreversible electrochemical reaction occurring in the cell. When a primary cell is discharged, it is discarded, and not recharged again like the secondary cell “rechargeable battery”. Dry cells and (most) alkaline batteries are examples of primary batteries.
- **Secondary Cells:** The second type of battery is rechargeable and is called a secondary battery. Examples of secondary batteries include nickel-cadmium (NiCd), lead acid, and lithium-ion batteries. A secondary battery can be used multiple times by recharging it.

Lead Acid Battery

- The lead-acid battery is a type of rechargeable battery first invented in 1859 by French physicist Gaston Planté. It is the first type of rechargeable battery ever created.
- Compared to modern rechargeable batteries, lead-acid batteries have relatively low energy density.
- A lead-acid battery consists of a negative electrode made of spongy or porous lead. The lead is porous to facilitate the formation and dissolution of lead. The positive electrode consists of lead oxide.
- The lead-acid battery consists of two electrodes submerged in an electrolyte of sulfuric acid. The positive electrode is made of grains of metallic lead oxide, while the negative electrode is attached to a grid of metallic lead.
- Lead-acid batteries are classified into two types: flooded and valve-regulated.

- Lead-acid batteries are currently used in uninterrupted power modules, electric grid, and automotive applications, including all hybrid and LIB-powered vehicles, as an independent 12-V supply to support starting, lighting, and ignition modules, as well as critical systems, under cold conditions and in the event of a high-voltage battery failure.
- A lead-acid battery system may cost hundreds or thousands of dollars less than a similarly-sized lithium-ion setup – lithium-ion batteries currently cost anywhere from \$5,000 to \$15,000 including installation, and this range can go higher or lower depending on the size of system you need.

Corrosion

- Corrosion is the process of deterioration of metal caused by the action of chemicals or electro-chemicals present in the surrounding atmosphere.
- It is the chemical or electrochemical destruction of metals and alloys in the presence of an environment to form undesirable compounds.
- Corrosion processes are the reactions of metals with environmental organisms.
- It is the process of a body gradually deteriorating due to the attack of air gases on the metal's surface.
- Corrosion is a major problem especially in the construction industry where various metals are used for structural purposes.

Types of Corrosion

- Erosion Corrosion: caused by mechanical abrasion due to the movement between corrosive fluids and the metal surface.
- Uniform Corrosion: corrosion readily identifiable by ordinary visual examination.
- Pitting Corrosion: corrosion readily identifiable by ordinary visual examination.
- Crevice Corrosion: corrosion readily identifiable by ordinary visual examination.
- Fillform Corrosion: corrosion readily identifiable by ordinary visual examination.
- Pack Rust Corrosion: corrosion readily identifiable by ordinary visual examination.
- Galvanic Corrosion: corrosion readily identifiable by ordinary visual examination.
- Lamellar Corrosion: corrosion readily identifiable by ordinary visual examination.
- Erosion Corrosion: corrosion requiring supplementary means of examination.
- Cavitation Corrosion: corrosion requiring supplementary means of examination.
- Fretting Corrosion: corrosion requiring supplementary means of examination.

tion.

- Intergranular Corrosion: corrosion requiring supplementary means of examination.

Cause of Corrosion

- The action of air, moisture, or a chemical reaction (such as an acid) on the metal surface.
- The attack of air gases on the metal's surface.
- The movement between corrosive fluids and the metal surface.

Corrosion Prevention

- Corrosion reactions can be used to advantage in technological processes such as pickling, etching, chemical, and electrochemical polishing and machining, etc.
- Any reaction of a metal with its environment must be regarded as a corrosion process irrespective of whether it is used to advantage or not.

Prevention and Control of Corrosion

Corrosion prevention and control (CPC) is a crucial aspect of maintaining physical assets, ranging from aircraft, ships, ground combat vehicles, and materiel, as well as wharves, buildings, and other infrastructure. These assets are subject to degradation due to corrosion, and CPC helps to reduce the effect of corrosion on sustainment costs while also improving materiel readiness.

CPC entails the characteristics of a system design to preclude or reduce corrosion, materials selection, non-destructive inspections for corrosion detection, coatings, finishes, cleaning materials and washings, repairs, and other maintenance activities .

The DoD Corrosion Prevention and Control Planning Guidebook for Military Systems and Equipment (MS&E) (i.e., CPC Planning Guidebook) helps Program Managers (PMs), Systems Engineers, Product Support Managers and other program staff develop and execute a comprehensive CPC approach.

Issues in Specific Industries

Power Generation

1. Environmental concerns: Power generation, particularly from fossil fuels, contributes to air and water pollution, greenhouse gas emissions, and climate change.
2. Reliability: Power outages and blackouts can have significant economic and social impacts.

3. Aging infrastructure: Many power plants and transmission lines are reaching the end of their useful lives and require significant investment to maintain or replace.
4. Regulatory challenges: The power generation industry is heavily regulated, and changes in regulations can have significant impacts on the industry.

Chemical Industry

1. Environmental concerns: The chemical industry produces and uses hazardous substances that can have negative impacts on the environment and human health.
2. Safety: Chemical plants and processes can be dangerous, and accidents can have significant consequences.
3. Regulatory challenges: The chemical industry is heavily regulated, and changes in regulations can have significant impacts on the industry.
4. Global competition: The chemical industry is highly competitive, and companies must constantly innovate and improve efficiency to remain competitive.

Processing Industry, Oil & Gas Industry and Pulp & Paper Industries

Processing Industry

- The processing industry involves the transformation of raw materials into finished products.
- This industry includes a wide range of sectors such as food processing, textile processing, and chemical processing.
- The processing industry plays a crucial role in the economy by adding value to raw materials and creating employment opportunities.

Oil & Gas Industry

- The oil and gas industry is involved in the exploration, extraction, refining, and distribution of oil and gas.
- This industry is critical to the global economy as it provides the primary source of energy for transportation, heating, and electricity generation.
- The oil and gas industry is divided into three main sectors: upstream, midstream, and downstream.

Pulp & Paper Industries

- The pulp and paper industry is involved in the production of paper and paper products from wood and other raw materials.

- This industry plays a vital role in the global economy by providing paper products for communication, packaging, and hygiene.
- The pulp and paper industry is divided into two main sectors: pulp production and paper production.

Chemistry of Engineering Materials:

1. Engineering materials are materials that are used in the construction of machines, structures, and other technical systems.
2. These materials can be classified into four main categories: metals, ceramics, polymers, and composites.
3. Metals are materials that are characterized by their high strength, ductility, and conductivity. They are used in a wide range of applications, including structural materials, electrical conductors, and heat exchangers.
4. Ceramics are materials that are characterized by their high hardness, strength, and resistance to heat and chemical attack. They are used in applications such as high-temperature components, cutting tools, and abrasives.
5. Polymers are materials that are characterized by their low density, high flexibility, and resistance to chemical attack. They are used in applications such as packaging, insulation, and coatings.
6. Composites are materials that are made up of two or more different materials, which are combined to produce a material with improved properties. They are used in applications such as aerospace structures, sports equipment, and automotive components.
7. The chemistry of engineering materials is concerned with understanding the properties of these materials at the atomic and molecular level, and using this understanding to design new materials with improved properties.
8. This involves studying the structure and bonding of materials, as well as their reactivity and stability under different conditions.
9. Techniques such as X-ray diffraction, spectroscopy, and microscopy are used to study the structure and properties of engineering materials.
10. The development of new engineering materials is an ongoing process, with researchers working to design materials with improved strength, durability, and performance.

Cement; Constituents, manufacturing, hardening and setting, deterioration of cement, Plaster

1. **Constituents:** Cement is a binding material that sets, hardens, and adheres to other materials to bind them together. It is manufactured through a closely controlled chemical combination of calcium, silicon, aluminum, iron, and other ingredients.
2. **Manufacturing:** The focus is on Portland cement, but attention is also given to other types, such as slag-containing cement and high-alumina ce-

ment. The historical development of cement, its manufacture from raw materials, its composition and properties, and the testing of those properties are surveyed.

3. **Hardening and Setting:** Cements of this kind are finely ground powders that, when mixed with water, set to a hard mass. Setting and hardening result from hydration, which is a chemical combination of the cement compounds with water that yields submicroscopic crystals or a gel-like material with a high surface area. Hydration, setting, and hardening of Portland cement are constrained by C3S, C3A. And all the admixtures that affect the hydration of C3S, C3A can change the performance of the hydration, the setting, and hardening of Portland cement .
4. **Deterioration of Cement:** Concrete can deteriorate for a variety of reasons, and concrete damage is often the result of a combination of factors. Set cement and concrete can suffer deterioration from attack by some natural or artificial chemical agents. The alumina compound is the most vulnerable to chemical attack in soils containing sulfate salts or in seawater, while the iron compound and the two calcium silicates are more resistant .
5. **Plaster:** Plaster is a building material used for the protective or decorative coating of walls and ceilings and for molding and casting decorative elements. It is made by heating gypsum to around 150°C. When the dry plaster powder is mixed with water, it re-forms into gypsum.

Paris (POP)

- According to recent estimates, the population of the city of Paris is 2,206,488, representing a small decline in population numbers from 2014 .
- However, the population of the surrounding suburbs is estimated to be around 10.5 million, which makes it the most populous urban area in the European Union .
- Paris's 2023 population is now estimated at 12,064,575 .
- In 1950, the population of Paris was 6,283,018 .
- Paris has grown by 68,610 in the last year, which represents a 0.57% annual change .
- These population estimates and projections come from the latest revision of the UN World Urbanization Prospects .
- In the past 70 years, Paris's population has almost doubled in size from 6.2 million people in 1950 .
- Paris has now become the most populous urban area in all of Europe, with a population density of 21,000 people per square kilometer .
- Paris owes most of its population growth to the birth of children between the years of 1999 to 2009 .

Unit-4: 8

1. I'm sorry, but I need more information to provide you with relevant content. Can you please specify the subject and context of Unit-4 and what the number 8 refers to?
2. Is this for a specific course or exam?
3. Any additional information you can provide will help me generate the most relevant and useful content for you.

Water Technology

Sources of Water

Water is a vital resource for life and is obtained from various sources. Some of the common sources of water are: 1. Surface water: This includes water from sources such as rivers, lakes, and reservoirs. 2. Groundwater: This is water that is found underground in aquifers and is obtained through wells and boreholes. 3. Rainwater: This is water that is collected from precipitation and can be harvested for use. 4. Sea water: This is water from the ocean and can be made potable through desalination.

Impurities in Water

Water from natural sources is rarely pure and often contains various impurities. Some common impurities found in water are: 1. Physical impurities: These include suspended particles such as sand, clay, and silt. 2. Chemical impurities: These include dissolved minerals such as calcium and magnesium, which can cause hardness in water. 3. Biological impurities: These include microorganisms such as bacteria, viruses, and algae.

Hardness of Water

Hardness in water is caused by the presence of dissolved minerals such as calcium and magnesium. There are two types of hardness in water: 1. Temporary hardness: This is caused by the presence of bicarbonates of calcium and magnesium and can be removed by boiling the water. 2. Permanent hardness: This is caused by the presence of sulfates and chlorides of calcium and magnesium and cannot be removed by boiling. It can be removed by methods such as ion exchange or reverse osmosis.

Boiler Troubles

Boiler troubles refer to problems that can occur in boilers due to the presence of impurities in the water used. Some common boiler troubles are: 1. Scaling: This is the deposition of minerals on the inner surfaces of the boiler, which can reduce heat transfer and efficiency. 2. Corrosion: This is the deterioration of the boiler material due to chemical reactions with the water and impurities. 3.

Priming and foaming: This is the carryover of water with the steam, which can cause problems in the steam system.

Techniques for Water Softening

Water softening is the process of removing impurities from water that can cause hardness. These impurities include calcium and magnesium ions, which can cause problems in industrial and domestic settings. There are several techniques for water softening, including Lime-Soda, Zeolite, Ion Exchange, and Reverse Osmosis.

1. **Lime-Soda:** The lime-soda method of water softening involves the use of chemicals such as calcium hydroxide (slaked lime) and sodium carbonate (soda ash) to remove impurities from water. This method must be followed by sedimentation and filtration in order to remove the precipitates.
2. **Zeolite:** The zeolite process for water softening has become a commercial success because zeolite can be easily regenerated. When hard water containing calcium and magnesium ions is passed through a bed of sodium zeolite, the sodium ions are replaced by the calcium and magnesium ions.
3. **Ion Exchange:** Ion exchange softening is applied to water high in non-carbonated hardness and the total hardness does not exceed 350 mg/L. This method involves the use of ion exchange resins to remove impurities from water.
4. **Reverse Osmosis:** Reverse osmosis softening involves the use of a semipermeable membrane to remove impurities from water. This method is effective at removing a wide range of impurities, including calcium and magnesium ions.

These are some of the techniques used for water softening. Each technique has its own advantages and disadvantages, and the choice of technique will depend on the specific needs and requirements of the user.

Hardness and Alkalinity

Hardness and alkalinity are two important parameters in water chemistry. Hardness refers to the concentration of calcium and magnesium ions in water, while alkalinity refers to the ability of water to neutralize acids.

Determination of Hardness

Hardness can be determined using several methods, including:

1. **EDTA titration:** This method involves titrating a water sample with a standard solution of ethylenediaminetetraacetic acid (EDTA) using a suitable indicator.

2. Atomic absorption spectroscopy (AAS): This method involves measuring the absorption of light by calcium and magnesium ions in a water sample.
3. Inductively coupled plasma optical emission spectroscopy (ICP-OES): This method involves measuring the emission of light by calcium and magnesium ions in a water sample.

Determination of Alkalinity

Alkalinity can be determined using several methods, including:

1. Titration with a standard acid: This method involves titrating a water sample with a standard solution of a strong acid, such as hydrochloric acid (HCl), using a suitable indicator.
2. Gran titration: This method involves titrating a water sample with a standard solution of a strong acid and plotting the titration curve to determine the alkalinity.
3. pH meter: This method involves measuring the pH of a water sample and using the pH value to calculate the alkalinity.

Numerical Problems

1. A water sample has a hardness of 150 mg/L as CaCO_3 . Calculate the concentration of calcium ions in the water sample in mg/L.

Solution: The molar mass of CaCO_3 is 100 g/mol, and the molar mass of Ca^{2+} is 40 g/mol. Therefore, the concentration of Ca^{2+} in the water sample is $(150 \text{ mg/L}) \times (40 \text{ g/mol}) / (100 \text{ g/mol}) = 60 \text{ mg/L}$.

2. A water sample has an alkalinity of 100 mg/L as CaCO_3 . Calculate the concentration of bicarbonate ions in the water sample in mg/L.

Solution: The molar mass of CaCO_3 is 100 g/mol, and the molar mass of HCO_3^- is 61 g/mol. Therefore, the concentration of HCO_3^- in the water sample is $(100 \text{ mg/L}) \times (61 \text{ g/mol}) / (100 \text{ g/mol}) = 61 \text{ mg/L}$.

Fuels and Combustion

Definition

- A fuel is a substance that is burned to produce heat or power.
- Combustion is the process of burning a fuel in the presence of oxygen to release energy in the form of heat and light.

Classification

Fuels can be classified into three main categories: 1. Solid fuels: such as coal, wood, and charcoal. 2. Liquid fuels: such as gasoline, diesel, and kerosene. 3. Gaseous fuels: such as natural gas, propane, and hydrogen.

Characteristics of a good fuel

A good fuel should have the following characteristics: 1. High calorific value: The fuel should release a large amount of heat per unit mass or volume when burned. 2. Easy to ignite: The fuel should ignite easily and burn steadily. 3. Clean burning: The fuel should burn cleanly, producing minimal pollutants and leaving little or no residue. 4. Safe to handle and store: The fuel should be safe to handle and store, with a low risk of accidental ignition or explosion. 5. Readily available: The fuel should be readily available and easy to obtain. 6. Economical: The fuel should be cost-effective, providing a high amount of energy for a reasonable price.

Calorific value

- The calorific value of a fuel is the amount of heat released per unit mass or volume of the fuel when it is completely burned.
- It is usually expressed in units of energy per unit mass, such as joules per gram (J/g) or calories per gram (cal/g).
- The higher the calorific value of a fuel, the more heat it releases when burned, and the more efficient it is as a source of energy.

Values, Gross & Net calorific value, Determination of calorific value by Bomb Calorimeter

- **Values** refer to the principles or standards of behavior that are considered important or worthwhile by an individual or society.
- **Calorific value** is the amount of heat energy released when a substance is burned completely.
- **Gross calorific value (GCV)** is the amount of heat released by the complete combustion of a unit of fuel, including the heat of vaporization of any water produced during combustion.
- **Net calorific value (NCV)** is the amount of heat released by the complete combustion of a unit of fuel, excluding the heat of vaporization of any water produced during combustion.
- The difference between GCV and NCV is the heat of vaporization of the water produced during combustion.
- **Bomb calorimeter** is a device used to determine the calorific value of a substance. It consists of a strong container (the bomb) in which the substance is burned, surrounded by a water bath to absorb the heat released during combustion.
- To determine the calorific value of a substance using a bomb calorimeter, the substance is weighed and placed in the bomb, which is then filled with oxygen and sealed. The bomb is placed in the water bath and the substance is ignited. The heat released during combustion is absorbed by the water, causing its temperature to rise. The temperature rise is

measured and used to calculate the calorific value of the substance.

Theoretical calculation of calorific value by Dulong's method, Ranking of Coal, Analysis of coal

Theoretical calculation of calorific value by Dulong's method

- Dulong's formula is used to calculate the calorific value of coal from its ultimate analysis.
- The formula is based on the proposition that the heat of combustion of an organic compound is nearly equal to the heats of combustion of the elements in it, multiplied by their percentage content in the compound in question.
- A new correlation developed to calculate calorific (heating) value of coal and char from their elemental composition is significantly better than other similar correlations.

Ranking of Coal

- Coal is ranked based on its degree of transformation from the original plant material to carbon.
- The ranks of coal, from lowest to highest, are lignite, sub-bituminous, bituminous, and anthracite.
- The rank of coal is determined by its carbon content, volatile matter, and heating value.

Analysis of coal

- Coal analysis is the process of determining the composition and properties of coal.
- The analysis can include measurements of the coal's moisture, volatile matter, ash, sulfur, and carbon content.
- The results of the analysis can be used to determine the coal's rank, its suitability for a particular use, and its environmental impact.

Proximate and Ultimate Analysis Method

Proximate and ultimate analysis are two methods used to determine the chemical composition of solid fuels, such as coal and biomass.

Proximate Analysis

Proximate analysis provides the percentage of the material that burns in a gaseous state (volatile matter), in the solid state (fixed carbon), and the per-

centage of inorganic waste material (ash). This analysis is of fundamental importance for biomass energy use.

Ultimate Analysis

Ultimate analysis, on the other hand, is based on the analysis of components of the gas phase that is formed during combustion of the studied samples in oxygen. This analysis typically involves the quantification of elements such as carbon, hydrogen, nitrogen, oxygen, and sulfur.

Numerical Problems

Numerical problems involving proximate and ultimate analysis may involve calculations based on the data obtained from these analyses. For example, a problem may involve determining the percentage of volatile matter, fixed carbon, and ash in a sample of coal based on the results of a proximate analysis.

Chemistry of Biogas

Biogas is a renewable energy source produced from the anaerobic digestion of organic matter. The chemical composition of biogas varies depending on the feedstock used and the conditions of the digestion process. However, the main components of biogas are typically methane (CH_4) and carbon dioxide (CO_2), with smaller amounts of other gases such as hydrogen sulfide (H_2S) and nitrogen (N_2).

Production from Organic Waste Materials and their Environmental Impact on Society

Organic waste materials can be used for production in a number of ways, including composting, anaerobic digestion, and energy recovery. These methods can help reduce the environmental impact of organic waste by reducing greenhouse gas emissions, conserving resources, and improving soil health.

- **Composting:** Composting is the process of breaking down organic waste materials into a nutrient-rich soil amendment. This process reduces the amount of methane emissions generated by organic waste in landfills, as composting significantly reduces these emissions. Compost can also reduce the need for chemical fertilizers, as it provides nutrients to the soil.
- **Anaerobic Digestion:** Anaerobic digestion is the process of breaking down organic waste materials in the absence of oxygen. This process produces biogas, which can be used as a renewable energy source. Anaerobic digestion can help reduce greenhouse gas emissions by capturing methane that would otherwise be released into the atmosphere.

- **Energy Recovery:** Energy recovery from organic waste materials involves converting the waste into a fuel source, such as through incineration or gasification. This process can help reduce greenhouse gas emissions by providing a renewable energy source and reducing the amount of waste sent to landfills.

However, it is important to note that the disposal of organic waste in landfills can have a significant environmental impact. Organic waste in landfills generates methane, a potent greenhouse gas. Landfills are one of the top emitters of man-made methane emissions in the United States. Methane is a greenhouse gas with a global warming potential approximately 85 times higher than carbon dioxide over a 20-year time period.

In conclusion, the production from organic waste materials can have a positive environmental impact by reducing greenhouse gas emissions, conserving resources, and improving soil health. However, the disposal of organic waste in landfills can have a significant negative impact on the environment due to the generation of methane emissions.

Unit-5: 8

1. Point 1
2. Point 2
3. Point 3
4. Point 4
5. Point 5
6. Point 6
7. Point 7
8. Point 8

Materials Chemistry:

Materials Chemistry is a branch of chemistry that deals with the study of the chemical and physical properties of materials. It involves the design, synthesis, and characterization of new materials with specific properties and applications.

Some key points to consider in Materials Chemistry are:

- Materials can be classified into different categories such as metals, ceramics, polymers, and composites.
- The properties of materials are determined by their chemical composition and structure.
- Materials chemists use various techniques to synthesize new materials, such as sol-gel processing, chemical vapor deposition, and self-assembly.
- Characterization techniques such as X-ray diffraction, electron microscopy, and spectroscopy are used to study the structure and properties of materials.
- Materials Chemistry has applications in various fields such as electronics, energy, medicine, and transportation.

Polymers; Classification, Polymerization processes, Thermosetting and Thermoplastic

Polymers are large molecules made up of repeating subunits called monomers. They can be classified based on their origin, structure, and composition.

1. Classification of Polymers:

- **Based on Origin:** Polymers can be classified as natural or synthetic. Natural polymers are found in nature, such as cellulose, starch, and proteins. Synthetic polymers are man-made, such as plastics, synthetic fibers, and rubber.
- **Based on Structure:** Polymers can be classified as linear, branched, or cross-linked. Linear polymers have a simple, long-chain structure. Branched polymers have side chains attached to the main chain. Cross-linked polymers have covalent bonds between the chains.
- **Based on Composition:** Polymers can be classified as homopolymers or copolymers. Homopolymers are made up of only one type of monomer. Copolymers are made up of two or more different types of monomers.

2. Polymerization Processes:

- **Addition Polymerization:** In this process, monomers with unsaturated bonds (double or triple bonds) react to form a polymer. The unsaturated bonds break and new bonds are formed between the monomers, creating a long chain.
- **Condensation Polymerization:** In this process, monomers react with each other to form a polymer and a small molecule, such as water, is eliminated. This process is also called step-growth polymerization.

3. Thermosetting and Thermoplastic:

- **Thermosetting Polymers:** These polymers are formed by a curing process that involves the formation of cross-links between the polymer chains. Once cured, they cannot be melted or reshaped. Examples include epoxy, phenolic, and melamine.
- **Thermoplastic Polymers:** These polymers can be melted and reshaped multiple times. They do not have cross-links between the polymer chains. Examples include polyethylene, polypropylene, and polyvinyl chloride.

Polymers, Polymer Blends and Composites, Conducting and Biodegradable polymers

- **Polymers** are large molecules composed of repeating structural units connected by covalent chemical bonds.
- **Polymer Blends** are mixtures of two or more polymers that are blended together to create a new material with different physical properties.

- **Composites** are materials made from two or more constituent materials with significantly different physical or chemical properties that, when combined, produce a material with characteristics different from the individual components.
- **Conducting Polymers** are organic polymers that conduct electricity.
- **Biodegradable Polymers** are polymers that can be decomposed by bacteria or other living organisms.

Biodegradable polymers are modified utilizing several methods such as blending and composites forming, which lead to new materials with unique properties including high performance, low cost, and good processability. Conducting polymers have been widely used in biomedical applications such as biosensors and tissue engineering but their non-degradability still poses a limitation. Therefore, great attention has been directed toward the recently developed degradable and electrically conductive polymers (DECPs).

Preparation, properties, industrial applications of Teflon, Lucite, Bakelite, Kelvar, Dacron

Teflon

Teflon is a brand name for the chemical compound polytetrafluoroethylene (PTFE). It is a thermally stable material that can withstand high temperatures. Its chemical properties make it ideal for use in various real-world applications, including electrical appliances, manufacturing, and the textile industry, where it is used in the production of many clothing items .

Properties of Teflon

- It is a white solid compound at room temperature.
- Its density is about 2200 kg/m³ or 2.2 g/cm³.
- Its melting point is 600 K.
- It is a chemically resistant compound, with only alkali metals being able to affect it.
- It shows good resistance to heat and low temperatures.
- It has a low water absorption capacity .

Bakelite

Bakelite is a type of plastic that is made from phenol and formaldehyde. It is prepared through several steps, including the formation of ortho and para hydroxy benzyl alcohols when phenol and formaldehyde react. Novolac is then created by removing a water molecule during the polymerization process of hydroxy benzyl alcohol, which is formed from phenol and formaldehyde in an acid medium with HCl or ZnCl₂ .

Properties of Bakelite

- Bakelite is a good insulator and is used in the manufacture of non-conducting parts of radio and electric devices such as sockets, wire insulation, switches, and automobile distribution caps.
- It is also used in the manufacture of clocks, buttons, washing machines, toys, kitchenware, and other items .

Thiokol, Nylon, Buna-N and Buna-S and their environmental impact on society, Speciality

- **Thiokol:** I'm sorry, but I couldn't find any information about the environmental impact of Thiokol.
- **Nylon:** Nylon is a synthetic fiber derived from petrochemicals, which are non-biodegradable, resulting in environmental degradation. The production of nylon is associated with the same negative environmental impacts as fossil fuels, including exacerbating the climate crisis with greenhouse gas emissions. Nylon clothing also contributes to microfiber pollution .
- **Buna-N:** Buna-N, also known as NBR rubber, is the seal industry's most widely used and economical elastomer. It displays excellent resistance to petroleum-based oils, fuels, water, alcohols, silicone greases, and hydraulic fluids . However, one of the major constraints to market growth in Europe and North America is environmental regulation and concern for the environmental impact associated with these products and their manufacture. For example, butadiene has been listed as a toxic substance under the Canadian Environmental Protection Act .
- **Buna-S:** Buna-S is similar to GR-S, a technology that proved crucial to solving the rubber crisis facing the United States during WWII . Buna® brand provides the automotive sector with a multilateral range of modern butadiene rubber products playing key roles in improving grip, resilience, and lowering rolling resistance. All these improvements go a long way toward helping to reduce CO2 emissions and positively impacting the planet .

Polymers

- Polymers are substances composed of macromolecules. A macromolecule is a molecule of high relative molecular mass, the structure of which essentially comprises the multiple repetition of units derived, actually or conceptually, from molecules of low relative molecular mass.
- Polymers range from familiar synthetic plastics such as polystyrene to natural biopolymers such as DNA and proteins that are fundamental to biological structure and function.

- Polymers, both natural and synthetic, are created via polymerization of many small molecules, known as monomers.
- Polymers make up many of the materials in living organisms, including, for example, proteins, cellulose, and nucleic acids.

Organometallic Compounds: General methods of preparation and applications of

Organometallic compounds are chemical compounds that contain at least one bond between a carbon atom of an organic molecule and a metal. These compounds have a wide range of applications in various fields, including catalysis, medicine, and materials science.

General methods of preparation:

1. Direct reaction of a metal with an organic compound: This method involves the direct reaction of a metal with an organic compound, such as an alkane or alkene, to form an organometallic compound.
2. Reaction of a metal with an organic halide: This method involves the reaction of a metal with an organic halide, such as an alkyl halide or aryl halide, to form an organometallic compound.
3. Reaction of a metal with an organometallic compound: This method involves the reaction of a metal with an organometallic compound, such as a Grignard reagent or an organolithium compound, to form a new organometallic compound.

Applications of organometallic compounds:

1. Catalysis: Organometallic compounds are widely used as catalysts in various chemical reactions, including polymerization, hydrogenation, and oxidation reactions.
2. Medicine: Organometallic compounds have been found to have various medicinal properties, including anti-cancer and anti-inflammatory properties.
3. Materials science: Organometallic compounds are used in the synthesis of various materials, including polymers, ceramics, and semiconductors.

Organometallic Compounds (RMgX and LiAlH_4)

Organometallic compounds are compounds that contain a metal-carbon bond. Two important types of organometallic compounds are RMgX and LiAlH_4 .

RMgX

RMgX , where X is a halide, is known as a Grignard reagent. Grignard reagents are named after Victor Grignard. They are usually written as R-Mg-X, but

in fact, the magnesium(II) center is tetrahedral when dissolved in Lewis basic solvents.

Grignard reagents are important due to their use in organic synthesis and as synthons for a range of organometallic compounds. They react with organic carbonyls (aldehydes, ketones, and esters) to yield the alcohol on hydrolysis. This synthetic route is useful for the formation of primary, secondary, and terminal alcohols.

LiAlH₄

LiAlH₄ is an organometallic compound that is commonly used as a reducing agent in organic synthesis. It is a powerful reducing agent that can reduce a wide range of functional groups, including aldehydes, ketones, esters, and carboxylic acids.

LiAlH₄ is a tetrahydridoaluminate complex that is formed by the reaction of lithium hydride with aluminum chloride. It is a white solid that is soluble in ether and tetrahydrofuran (THF) but reacts violently with water, releasing hydrogen gas.

In summary, organometallic compounds RMgX and LiAlH₄ are important reagents in organic synthesis due to their ability to react with a wide range of functional groups to form new compounds.

Course Outcomes:

- Define and explain the key concepts and principles of the subject.
- Demonstrate an understanding of the subject's theories and their applications.
- Analyze and evaluate information, arguments, and evidence related to the subject.
- Apply the subject's concepts and principles to solve problems and make decisions.
- Communicate effectively in written and oral forms about the subject.
- Work effectively in a team to achieve common goals related to the subject.
- Demonstrate ethical and professional behavior in the context of the subject.
- Engage in lifelong learning and professional development related to the subject.

Upon completion of the course the student should be able to:

1. Demonstrate a comprehensive understanding of the course material.
2. Apply the concepts and theories learned in the course to real-world situations.
3. Analyze and evaluate information critically and effectively.
4. Communicate effectively in both written and oral forms.

5. Work collaboratively with others to achieve common goals.
6. Demonstrate ethical and professional behavior.
7. Utilize technology effectively to enhance learning and problem-solving.
8. Engage in lifelong learning and professional development.

Units Course Outcomes Bloom's

1. **Units:** Units are a standard of measurement used to express physical quantities. They provide a common language for scientists and engineers to communicate their findings and ideas.
2. **Course Outcomes:** Course outcomes are the specific knowledge, skills, and abilities that students are expected to achieve by the end of a course. They are used to assess the effectiveness of the course and to guide future improvements.
3. **Bloom's Taxonomy:** Bloom's Taxonomy is a framework for categorizing educational goals and objectives. It is used to design learning activities and assessments that promote higher-order thinking skills. The taxonomy is organized into six levels: remembering, understanding, applying, analyzing, evaluating, and creating.

Level

- A level is a tool used to determine if a surface is horizontal (level) or vertical (plumb).
- Levels can be used in construction, carpentry, surveying, and many other applications.
- There are several types of levels, including spirit levels, laser levels, and water levels.
- Spirit levels use a liquid-filled vial with an air bubble to indicate levelness.
- Laser levels project a beam of light to create a straight line on a surface.
- Water levels use the principle that water will always find its own level to determine levelness.
- Levels can vary in size and accuracy, with larger and more precise levels being used for more demanding applications.
- It is important to use a level when installing or building structures to ensure that they are straight and level.

CO-1 Get an understanding of the theoretical principles of chemistry of molecular structure, bonding and

1. **Molecular Structure:** The arrangement of atoms in a molecule determines its shape and properties. Theories such as VSEPR (Valence Shell Electron Pair Repulsion) and hybridization are used to predict the geometry of molecules.

2. **Bonding:** Chemical bonds are the forces that hold atoms together in a molecule. There are several types of chemical bonds, including covalent, ionic, and metallic bonds. The type of bond formed depends on the electronegativity of the atoms involved.
3. **Theoretical Principles:** The principles of chemistry are based on the behavior of atoms, molecules, and ions. These principles include the laws of thermodynamics, the kinetic molecular theory, and the principles of chemical equilibrium.

These are some of the key concepts that are important for understanding the chemistry of molecular structure and bonding. It is important to have a strong foundation in these principles in order to understand more advanced topics in chemistry.

Properties, Chemistry of Advanced Materials

Liquid Crystals

- Liquid crystals are a state of matter that has properties between those of a conventional liquid and those of a solid crystal.
- They can flow like a liquid, but have the long-range order of a solid.
- Liquid crystals are used in displays, such as LCD screens, because they can be manipulated to control the passage of light.
- There are several types of liquid crystals, including nematic, smectic, and cholesteric.

Nanomaterials

- Nanomaterials are materials with at least one dimension in the nanometer range (1-100 nm).
- They have unique properties due to their small size, such as increased surface area and quantum effects.
- Nanomaterials are used in a variety of applications, including electronics, medicine, and energy production.
- Some common types of nanomaterials include nanoparticles, nanotubes, and nanowires.

Graphite

- Graphite is a form of carbon that consists of layers of carbon atoms arranged in a hexagonal lattice.
- It is a good conductor of heat and electricity and has a high melting point.
- Graphite is used in a variety of applications, including batteries, lubricants, and pencils.
- Graphite can be transformed into other forms of carbon, such as diamond and graphene, under certain conditions.

Fullerene

- Fullerenes are a class of carbon molecules that consist of carbon atoms arranged in a spherical or ellipsoidal shape.
- The most well-known fullerene is the buckminsterfullerene, or “buckyball,” which consists of 60 carbon atoms arranged in a soccer ball-like shape.
- Fullerenes have unique properties, such as high stability and the ability to encapsulate other molecules.
- They have potential applications in medicine, electronics, and materials science.

Principles of Green Chemistry

Green chemistry is the approach in chemical sciences that efficiently uses renewable raw materials, eliminates waste, and avoids the use of toxic and hazardous reagents and solvents in the manufacture and application of chemical products .

There are twelve basic principles of Green Chemistry that can be grouped into “Reducing Risk” and “Minimizing the Environmental Footprint” . Some of these principles include:

1. **Prevention of waste or by-products:** According to this principle of green chemistry, priority is given to the prevention of waste rather than cleaning up and treating waste after it has been generated .
2. **Atom Economy:** This principle focuses on designing synthetic methods to maximize the incorporation of all materials used in the process into the final product .
3. **Reducing the use of derivatives and protecting groups:** One of the key principles of green chemistry is to reduce the use of derivatives and protecting groups in the synthesis of target molecules. One of the best ways of doing this is the use of enzymes .

Green chemistry applies across the life cycle of a chemical product, including its design, manufacture, use, and ultimate disposal .

K3

- K3 is a term that can refer to several different things, depending on the context in which it is used.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a visa category for spouses of U.S. citizens.
- In mathematics, K3 could refer to a complete graph with three vertices.
- In music, K3 is a Belgian-Dutch girl group.
- In the context of technology, K3 could refer to a model of a product, such as a keyboard or a camera.

- Without more information, it is difficult to determine the specific meaning of K3 in this context.

CO-2 Apply the fundamental concepts of determination of structure with various spectral techniques

Spectral techniques are used to determine the structure of molecules by analyzing the interaction of electromagnetic radiation with matter. Some of the fundamental concepts of determination of structure with various spectral techniques are:

1. **Infrared (IR) Spectroscopy:** This technique measures the absorption of infrared radiation by a molecule. The absorption of infrared radiation causes changes in the vibrational energy of the molecule, which can be used to determine the functional groups present in the molecule.
2. **Nuclear Magnetic Resonance (NMR) Spectroscopy:** This technique measures the absorption of radiofrequency radiation by the nuclei of atoms in a molecule. The absorption of radiofrequency radiation causes changes in the spin state of the nuclei, which can be used to determine the chemical environment of the nuclei and the connectivity of atoms in the molecule.
3. **Ultraviolet-Visible (UV-Vis) Spectroscopy:** This technique measures the absorption of ultraviolet and visible radiation by a molecule. The absorption of ultraviolet and visible radiation causes changes in the electronic energy of the molecule, which can be used to determine the presence of conjugated systems and chromophores in the molecule.
4. **Mass Spectrometry (MS):** This technique measures the mass-to-charge ratio of ions produced by the molecule. The mass-to-charge ratio of the ions can be used to determine the molecular weight and the elemental composition of the molecule.

These are some of the fundamental concepts of determination of structure with various spectral techniques. Each technique provides different information about the molecule, and a combination of these techniques can be used to determine the complete structure of the molecule.

Stereochemistry

Stereochemistry is a subdiscipline of chemistry that involves the study of the relative spatial arrangement of atoms that form the structure of molecules and their manipulation. It is also known as 3D chemistry, where the prefix “stereo” means “three-dimensionality”.

The study of stereochemistry focuses on the relationships between stereoisomers, which by definition have the same molecular formula and sequence of bonded

atoms. A property of molecules that is dependent on their handedness is called chirality.

Some important concepts in stereochemistry include: - Chirality - Enantiomers - Diastereomers - Assigning configurations using the R,S system - Optical activity - Fischer projections

Stereochemistry is a systematic presentation of a specific field of science and technology that traditionally requires a short preliminary excursion into history.

K4

1. K4 is a term that can refer to several different things, depending on the context in which it is used.
2. In mathematics, K4 is the complete graph on four vertices, also known as a tetrahedron.
3. In cryptography, K4 is the fourth part of the Kryptos sculpture at the CIA headquarters, which contains an encrypted message that has not yet been fully deciphered.
4. In transportation, K4 may refer to a class of steam locomotives, a model of car, or a type of kayak.
5. In biology, K4 can refer to lysine 4, a specific location on a histone protein where modifications can occur that affect gene expression.
6. In geography, K4 is the name of a mountain in the Karakoram range in Pakistan.

CO-3 Utilize the theory of construction of electrodes, batteries and fuel cells in redesigning new engineering

1. **Electrodes:** Electrodes are conductive materials that allow the flow of electric current in and out of a cell or other device. They are typically made of metals or other conductive materials and are used in batteries, fuel cells, and other electrochemical devices.
2. **Batteries:** Batteries are devices that store electrical energy in the form of chemical energy and release it as needed. They consist of one or more electrochemical cells, each of which contains two electrodes and an electrolyte.
3. **Fuel Cells:** Fuel cells are electrochemical devices that convert the chemical energy of a fuel directly into electricity. They consist of an anode, a cathode, and an electrolyte. The fuel is oxidized at the anode, releasing electrons that flow through an external circuit to the cathode, where they combine with oxygen to form water.
4. **Redesigning New Engineering:** The theory of construction of electrodes, batteries, and fuel cells can be utilized in redesigning new engineering by improving the efficiency, durability, and performance of these devices. This can be achieved by using new materials, optimizing the design, and incorporating new technologies.

Products and Corrosion

Corrosion is the process of deterioration of materials due to chemical or electrochemical reactions with their environment. It is a common problem in many industries, including construction, transportation, and manufacturing. There are several reasons for corrosion and various methods to control it.

Reasons for Corrosion

1. **Chemical reactions:** Corrosion can occur due to chemical reactions between the material and its environment. For example, iron reacts with oxygen and water to form rust.
2. **Electrochemical reactions:** Corrosion can also occur due to electrochemical reactions, where the material acts as an anode and corrodes, while another material acts as a cathode and is protected.
3. **Environmental factors:** Environmental factors such as humidity, temperature, and the presence of pollutants can accelerate the corrosion process.
4. **Material properties:** The properties of the material, such as its composition and microstructure, can also affect its susceptibility to corrosion.

Methods to Control Corrosion

1. **Material selection:** Choosing materials that are resistant to corrosion in the specific environment can help prevent corrosion.
2. **Coatings:** Applying coatings, such as paint or galvanization, can protect the material from the environment and prevent corrosion.
3. **Cathodic protection:** Cathodic protection is an electrochemical method to control corrosion, where a sacrificial anode is used to protect the material from corrosion.
4. **Environmental control:** Controlling the environment, such as reducing humidity or the presence of pollutants, can help prevent corrosion.

It is important to study the causes and methods to control corrosion in order to develop products that are durable and resistant to corrosion. This can help reduce maintenance costs and increase the lifespan of the products.

Understanding of Chemistry of Engineering materials (Cement)

Cement is an adhesive substance used in building and civil engineering construction. It is a finely ground powder that sets to a hard mass when mixed with water .

- **Chemical Composition:** Portland cement is made up of four main compounds: tricalcium silicate ($3\text{CaO} \cdot \text{SiO}_2$), dicalcium silicate ($2\text{CaO} \cdot$

SiO₂), tricalcium aluminate (3CaO · Al₂O₃), and a tetra-calcium aluminoferrite (4CaO · Al₂O₃Fe₂O₃) .

- **Types of Cement:** Cement is chiefly of two kinds based on the way it is set and hardened: hydraulic cement, which hardens due to the addition of water, and non-hydraulic cement, which is hardened by carbonation with the carbon present in the air, so it cannot be used underwater .
- **Uses:** Cements may be used alone (i.e., “neat,” as grouting materials), but the normal use is in mortar and concrete in which the cement is mixed with inert material known as aggregate. Mortar is cement mixed with sand or crushed stone that must be less than approximately 5 mm (0.2 inch) in size .
- **Future Developments:** With a better understanding of cement chemistry, scientists could potentially alter production or change ingredients so that concrete has less of an impact on emissions, or add ingredients that are capable of actively absorbing carbon dioxide .

K3

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- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a visa category for spouses of U.S. citizens.
- In mathematics, K3 surfaces are a type of algebraic surface.
- In music, K3 is a Belgian-Dutch girl group.
- In technology, K3 is a model of amateur radio transceiver produced by Elecraft.
- Without more context, it is difficult to determine which specific meaning of K3 is being referred to.

CO-4 Develop understanding of the sources, impurities and hardness of water, apply the concepts of

1. **Sources of water:** Water can be obtained from various sources such as rivers, lakes, oceans, groundwater, and rainwater.
2. **Impurities in water:** Water from natural sources may contain impurities such as dissolved minerals, organic matter, and microorganisms. These impurities can affect the taste, odor, and appearance of water, as well as its suitability for certain uses.
3. **Hardness of water:** Hardness refers to the presence of dissolved minerals, primarily calcium and magnesium, in water. Hard water can cause problems such as scaling in pipes and reduced effectiveness of soaps and detergents.
4. **Concepts of water treatment:** Water treatment involves the removal of impurities from water to make it suitable for its intended use. This can

be achieved through various methods such as filtration, sedimentation, and disinfection.

Determination of Calorific Values and Analysis of Coal

Calorific value is a measure of the amount of energy produced from a unit weight of coal when it is combusted in oxygen. A measured sample of coal is completely combusted in a bomb calorimeter, which is a device for measuring heat. This method is described in the ASTM method D5865-12.

There are two categories of calorimeters for coal analysis: adiabatic and isoperibolic. Standard bomb calorimeters can be used to determine the gross calorific value and are described in the ISO 1928 and ASTM D5865 standard.

Elemental or ultimate analysis encompasses the quantitative determination of carbon, hydrogen, nitrogen, sulfur, and oxygen within the coal. Additionally, specific physical and mechanical properties of coal and particular carbonization properties are also determined.

Moisture content is an important parameter in coal analysis. It is needed for determining the calorific (heating) value and handling properties of a coal. The calorific value mainly comes from the combustion of organic components (C, H, N). In addition, moisture and ash in coal will absorb part of the heat during combustion.

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CO-5 Develop the understanding of Chemical structure of polymers and its effect on their various properties

1. Polymers are large molecules made up of repeating units called monomers.
2. The chemical structure of a polymer refers to the arrangement of its monomer units and the type of bonds that hold them together.
3. The chemical structure of a polymer can affect its properties such as strength, elasticity, melting point, and solubility.

4. For example, polymers with strong covalent bonds between their monomer units tend to have high melting points and are resistant to solvents.
5. On the other hand, polymers with weak intermolecular forces between their monomer units tend to have low melting points and are soluble in certain solvents.
6. The arrangement of monomer units in a polymer can also affect its properties. For example, polymers with a linear arrangement of monomer units tend to be strong and rigid, while polymers with a branched arrangement tend to be more flexible.
7. The presence of functional groups in a polymer can also affect its properties. For example, the presence of polar functional groups can increase the solubility of a polymer in polar solvents.
8. In summary, the chemical structure of a polymer plays a crucial role in determining its properties and behavior. Understanding the relationship between the chemical structure and properties of polymers is essential for their effective use in various applications.

Applications of Polymers in Engineering and Chemistry

Polymers are incredibly diverse elements that represent such fields of engineering from avionics through biomedical applications, drug delivery system, biosensor devices, tissue engineering, cosmetics etc. The application of polymers and their subsequent composites is still advancing and increasing quickly due to their ease regarding manufacturing.

Some of the industries in which you would see the use and application of various polymeric materials and polymers themselves are: 1. In aircraft, aerospace, and sports equipment 2. 3D printing plastics 3. Biopolymers in molecular recognition 4. Polymers in holography 5. Organic polymers used in water purification 6. Printed circuit board substrates 7. Green Chemicals: Polymers and Biopolymers 8. Polymeric Biomolecules

Polymer chemists have designed and synthesized polymers that vary in hardness, flexibility, softening temperature, solubility in water, and biodegradability. They have produced polymeric materials that are as strong as steel yet lighter and more resistant to corrosion.

Sample research topics that span the journal's scope are polymer applications in energy storage and conversion, separations, membranes, adhesives, functional coatings, sensing, adaptive and reconfigurable materials, electronics, photonics, biomaterials, and nanocomposites.

Applicable in Industrial Process

1. **Automation:** Automation is the use of control systems and information technologies to reduce the need for human intervention in the production process. Automation can increase efficiency, reduce costs, and improve product quality.
2. **Quality Control:** Quality control is the process of ensuring that products meet the required standards of quality. This can be achieved through various methods such as inspection, testing, and statistical process control.
3. **Supply Chain Management:** Supply chain management is the coordination and management of activities involved in the production and delivery of goods and services. This includes sourcing raw materials, managing inventory, and coordinating logistics.
4. **Lean Manufacturing:** Lean manufacturing is a systematic approach to reducing waste and increasing efficiency in the production process. This can be achieved through techniques such as just-in-time production, continuous improvement, and standardization of work processes.
5. **Computer-Aided Design and Manufacturing:** Computer-aided design (CAD) and computer-aided manufacturing (CAM) are technologies that allow for the design and production of products using computer software. This can improve the speed and accuracy of the design and manufacturing process.
6. **Energy Efficiency:** Energy efficiency is the practice of reducing the amount of energy required to produce goods and services. This can be achieved through the use of energy-efficient technologies, process improvements, and energy management systems.
7. **Waste Management:** Waste management is the process of reducing, reusing, and recycling waste generated during the production process. This can reduce the environmental impact of industrial activities and improve the sustainability of the production process.
8. **Health and Safety:** Health and safety is the practice of ensuring that the production process is safe for workers and does not pose a risk to their health. This can be achieved through the use of personal protective equipment, training, and the implementation of safety procedures.
9. **Research and Development:** Research and development (R&D) is the process of developing new products, processes, and technologies. R&D can drive innovation and improve the competitiveness of industrial companies.
10. **Data Analysis:** Data analysis is the process of collecting, processing, and analyzing data to extract useful information. Data analysis can be used to improve decision-making, identify trends, and optimize the production process.

K3

- K3 is a common abbreviation that can refer to several different things.
- It can refer to the third highest mountain in the Karakoram range, also known as Broad Peak.
- It can also refer to a type of visa for spouses of U.S. citizens to enter the United States while waiting for their immigrant visa to be processed.
- In mathematics, K3 can refer to a complete graph with three vertices.
- In education, K3 can refer to the third year of primary education, typically for children aged 8-9 years old.
- In music, K3 is a Belgian-Dutch girl group.
- In transportation, K3 can refer to a class of steam locomotives used by the London and North Eastern Railway.
- In technology, K3 can refer to a model of amateur radio transceiver produced by Elecraft.

Reference Books:

- Reference books are a type of non-fiction book that provides factual information on a particular subject.
- They are often used for research and academic purposes.
- Some common types of reference books include dictionaries, encyclopedias, almanacs, and atlases.
- Reference books are usually organized in a way that makes it easy to find the information you need.
- They often include indexes, tables of contents, and other tools to help you navigate the content.
- Reference books are usually not meant to be read cover-to-cover, but rather to be consulted for specific information.
- Many reference books are updated regularly to ensure that the information they contain is accurate and up-to-date.
- Some reference books are available in digital format, making it easy to access the information you need from anywhere.
- When using a reference book for research, it is important to evaluate the credibility of the source and to cite the information properly in your work.

Engineering Chemistry by Rath & Singh, 2nd Edition, Cengage Learning India Pvt Ltd Delhi

- Engineering Chemistry is a textbook written by Rath and Singh.
- It is published by Cengage Learning India Pvt Ltd Delhi.
- The book is in its 2nd edition.
- It covers various topics in the field of engineering chemistry.
- The book is designed to provide a comprehensive understanding of the subject.
- It is suitable for students studying engineering chemistry as part of their curriculum.

- The book is written in a clear and concise manner, making it easy to understand.
- It includes numerous examples and illustrations to aid in the understanding of the concepts presented.
- The book is a valuable resource for students preparing for exams in the subject of engineering chemistry.

2. Engineering Chemistry by SS Dara, S Chand & Co Ltd

- “Engineering Chemistry” is a textbook written by S. S. Dara and published by S. Chand & Co Ltd.
- The book is written exclusively for students of all branches of engineering, keeping in view their professional requirements.
- It is a comprehensive book, divided into 26 chapters.
- The latest edition of the book is available for purchase.
- The book is also available for free download as a PDF file.

3. Engineering Chemistry by Jain & Jain, S.Chand & Comp, New Delhi

- Engineering Chemistry is a book written by Jain and Jain and published by Dhanpat Rai Publishing Company .
- It is one of the primary textbooks for the course CY100 Engineering Chemistry for Kerala Technological University (KTU) .
- The original textbook may cost up to Rs.425 .
- The 17th edition of the book has been made more informative and comprehensive .
- This book has been prepared specially for students, keeping in view the new syllabus of various reputed universities throughout the country .
- It is written exclusively for students of all branches of engineering, keeping in view their professional requirements .
- The book is divided into 26 chapters .
- The 16th edition of the book is also available in paperback format on Amazon.in .

4. Engineering Chemistry by K. Sesha Maheswaramma, Pearson

Engineering Chemistry is a textbook written by K. Sesha Maheswaramma and published by Pearson. The book covers the fundamental concepts of chemistry and their applications in engineering. Some of the topics covered in the book include:

1. Atomic Structure and Chemical Bonding
2. States of Matter
3. Chemical Thermodynamics
4. Electrochemistry
5. Chemical Kinetics

6. Surface Chemistry
7. Coordination Compounds
8. Organic Chemistry
9. Polymers
10. Environmental Chemistry

This book is intended for undergraduate students in engineering and provides a comprehensive introduction to the subject of chemistry in the context of engineering. It is written in a clear and concise manner and includes numerous examples and illustrations to aid in understanding the material. The book also includes end-of-chapter problems to help students test their knowledge and understanding of the concepts presented.

5. Engineering Chemistry by OG Palanna, Mc Graw Hill Education, New Delhi

- “Engineering Chemistry” is a textbook written by OG Palanna and published by Mc Graw Hill Education, New Delhi.
- The book is intended for undergraduate students in engineering and covers the fundamental concepts of chemistry as applied to engineering.
- The book is divided into several chapters, each covering a specific topic in the field of engineering chemistry.
- Some of the topics covered in the book include atomic structure, chemical bonding, thermodynamics, electrochemistry, and materials science.
- The book is written in a clear and concise manner, with numerous examples and illustrations to aid in understanding the concepts presented.
- Overall, “Engineering Chemistry” by OG Palanna is a valuable resource for students studying engineering and seeking to gain a deeper understanding of the role of chemistry in the field.

6. Engineering Chemistry by Shashi Chawala, Dhanpat Rai Publishing Comp, New Delhi

- Engineering Chemistry is a textbook written by Shashi Chawla and published by Dhanpat Rai Publishing Comp, New Delhi .
- The book covers the syllabus for the Engineering Chemistry course offered to first-year B.E/B.Tech students of various Indian universities .
- The book includes topics such as electrochemistry, environmental chemistry, organic reaction mechanism, etc .
- The book has received positive reviews and ratings, with an average rating of 4.18 on Goodreads .

7. University Chemistry by BH Mahan

- University Chemistry by BH Mahan is a comprehensive textbook for undergraduate students studying chemistry.

- The book covers fundamental concepts in chemistry, including atomic structure, chemical bonding, and chemical reactions.
- It also includes chapters on thermodynamics, kinetics, and electrochemistry.
- The book is known for its clear explanations and numerous examples and practice problems.
- It is widely used as a reference for students preparing for exams in chemistry.
- The book is written in a formal style and is well-organized, making it easy to navigate and find information.
- Overall, University Chemistry by BH Mahan is a valuable resource for students studying chemistry at the university level.

8. University Chemistry by CNR Rao

- University Chemistry is a book written by CNR Rao.
- CNR Rao is an Indian chemist who has worked mainly in solid-state and structural chemistry.
- The book is intended for undergraduate students studying chemistry.
- It covers the fundamental concepts of chemistry, including atomic structure, chemical bonding, and chemical reactions.
- The book also includes chapters on thermodynamics, electrochemistry, and chemical kinetics.
- It is written in a clear and concise manner, making it easy for students to understand the material.
- The book includes numerous examples and practice problems to help students apply the concepts they have learned.
- Overall, University Chemistry by CNR Rao is a valuable resource for students studying chemistry at the university level.